

Docker Desktop, Docker Compose eta Python 3 instalatzeko gida (Windows 11)

Dokumentu honetan Docker Desktop Windows 11 sistema eragilean instalatzeko beharrezko urratsak azaltzen dira. Instalazioa egin aurretik, sistema behar bezala prestatzea ezinbestekoa da; bereziki, BIOS/UEFIko birtualizazioa gaitzea eta WSL2 (Windows Subsystem for Linux 2) instalatzea.

1. Docker Desktop instalatzea

1.1. Aurrebaldintzak

Instalazioa hasi aurretik, egiaztatu honako baldintzak betetzen direla:

- Windows 11 (64 bit)
- Administrazioaile-baimenak
- Interneteko konexioa
- BIOS/UEFI bidez birtualizazioa gaitzeko aukera

1.2. Birtualizazioa gaitzea

Docker Desktop-ek hardware bidezko birtualizazioa behar du.

- **Birtualizazioa aktibatuta dagoen egiaztatzea**
 1. Sakatu **Ctrl + Shift + Esc** Task Manager irekitzeko.
 2. Joan **Performane (Rendimiento) > CPU** atalera.
 3. Begiratu **Virtualization (Virtualización)** eremua:
 - **Enabled (Habilitado)** bada, “*Virtual Machine Platform gaituta dagoen egiaztatzea*” puntura joan.
 - **Disabled (Deshabilitado)** bada, BIOS/UEFIra sartu beharko da:
 - Ordenagailua berrabiarazi.
 - Abioan fabrikatzailearen tekla sakatu (normalean **F2**, **F10**, **F12**, **DEL** edo **ESC**).
 - Bilatu aukera hauetako bat:
 - a. Intel Virtualization Technology (VT-x)
 - b. Intel VT-d
 - c. AMD-V
 - d. SVM Mode
 - Aukera **Enabled** moduan jarri.
 - Gorde aldaketak eta berrabiarazi ordenagailua.

- **Virtual Machine Platform gaituta dagoen egiaztatzea**

1. Ireki PowerShell administratzaile gisa.
2. Exekutatu komando hau:

```
dism.exe /online /get-features /format:table | findstr /i "Microsoft-Windows-Subsystem-Linux VirtualMachinePlatform"
```

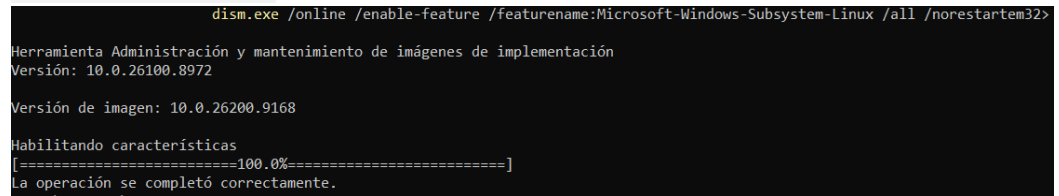
3. Exekuzioaren bukaeran, hau agertu beharko litzateke:

```
VirtualMachinePlatform | Habilitado  
Microsoft-Windows-Subsystem-Linux | Habilitado
```

- **Gaituta (Habilitado)** badaude, 1.3. puntura joan.
- **Ez badaude gaituta (Deshabilitado)**, hurrengo pausoak jarraitu:

- PowerShell-en, exekutatu komando hau:

```
dism.exe /online /enable-feature  
/featurename:Microsoft-Windows-Subsystem-Linux  
/all /norestart
```

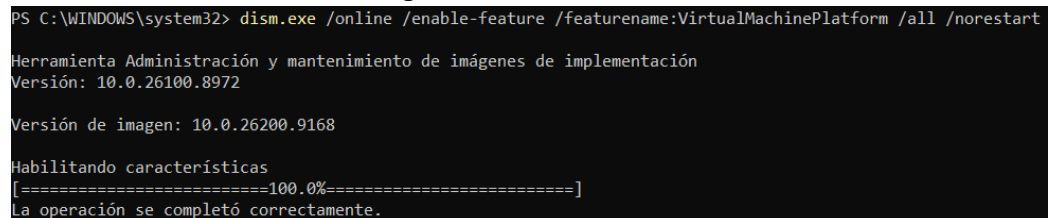


A screenshot of a PowerShell window showing the execution of the command `dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart`. The output shows the Windows Imaging and Deployment Tool version 10.0.26100.8972, the image version 10.0.26200.9168, and a progress bar indicating that the operation is 100% complete. The final message is "La operación se completó correctamente."

- Ondoren, exekutatu bigarren komando hau:

```
dism.exe /online /enable-feature  
/featurename:VirtualMachinePlatform /all  
/norestart
```

Amaitzean, honelako mezua agertu beharko litzateke:

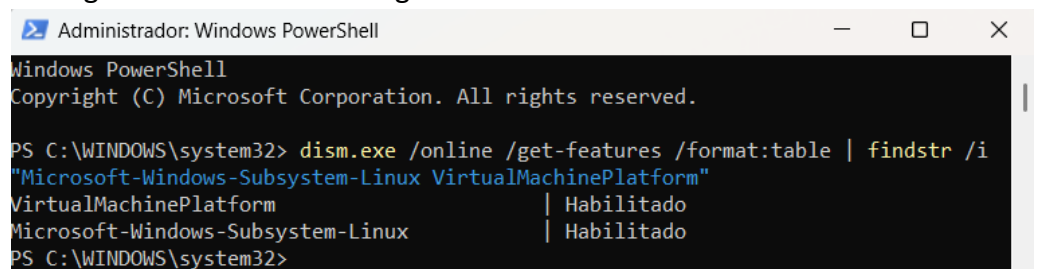


A screenshot of a PowerShell window showing the execution of the command `dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart`. The output shows the Windows Imaging and Deployment Tool version 10.0.26100.8972, the image version 10.0.26200.9168, and a progress bar indicating that the operation is 100% complete. The final message is "La operación se completó correctamente."

- Berrabiarazi Windows 11.
- Egiaztatu bi funtzioak aktibatuta daudela, berriro ere PowerShell-en komando hau exekutatuz:

```
dism.exe /online /get-features /format:table |  
findstr /i "Microsoft-Windows-Subsystem-Linux VirtualMachinePlatform"
```

Oraingoan honelako zerbait agertu beharko litzateke:



A screenshot of a PowerShell window titled "Administrador: Windows PowerShell". It shows the execution of the command `dism.exe /online /get-features /format:table | findstr /i "Microsoft-Windows-Subsystem-Linux VirtualMachinePlatform"`. The output displays a table with two rows: "VirtualMachinePlatform" and "Microsoft-Windows-Subsystem-Linux", both with the status "Habilitado".

1.3. WSL2 instalatzea

Docker Desktop-ek WSL2 erabiltzen du Linux edukiontziak exekutatzeko.

- **PowerShell administratzaile moduan ireki**

- Hasierako menuan **PowerShell** bilatu.
- Aukeratu **Run as administrator**.

- **WSL instalatu**

Exekutatu komando hau:

```
wsl --install
```

Komando honek: WSL instalatzen du, WSL2 ezartzen du eta Ubuntu instalatzen du lehenetsitako banaketa gisa.

- Komandoa **ondo** exekutatzen bada, amaitzean ordenagailua berrabiarazi.
- Komando honek **errorea** ematen badu:
 - WSLren azken MSI paketea deskargatu GitHub Releases-etik:
https://github.com/microsoft/WSL/releases?utm_source=chatgpt.com
 - Bertan, azken bertsioan, bilatu **Assets** atala eta deskargatu zure arkitekturari dagokion .msi fitxategia (zure ordenagailua Intel/AMD bada, normalean x64 izango da).
 - Deskargatu ondoren, egin klik bikoitza .msi fitxategian eta egin instalazioa.
 - Berrabiarazi.

- **WSL2 dela egiaztatu**

PowerShell-en exekutatu:

```
wsl --status
```

Espero den emaitza: *Default Version: 2*

- **Instalatutako banaketak ikusi**

PowerShell-en exekutatu:

```
wsl --list --verbose
```

Adibidez:

NAME	STATE	VERSION
Ubuntu	Running	2

VERSION zutabea 2 agertu behar da.

1.4. Deskargatu Docker Desktop

1. Ireki nabigatzailea.
2. Joan Docker-en webgune ofizialeko helbide honetara:
<https://www.docker.com/products/docker-desktop/>
3. Sakatu **Download Docker Desktop** → **Download for Windows – AMD64**.
4. Itxaron deskarga amaitu arte.

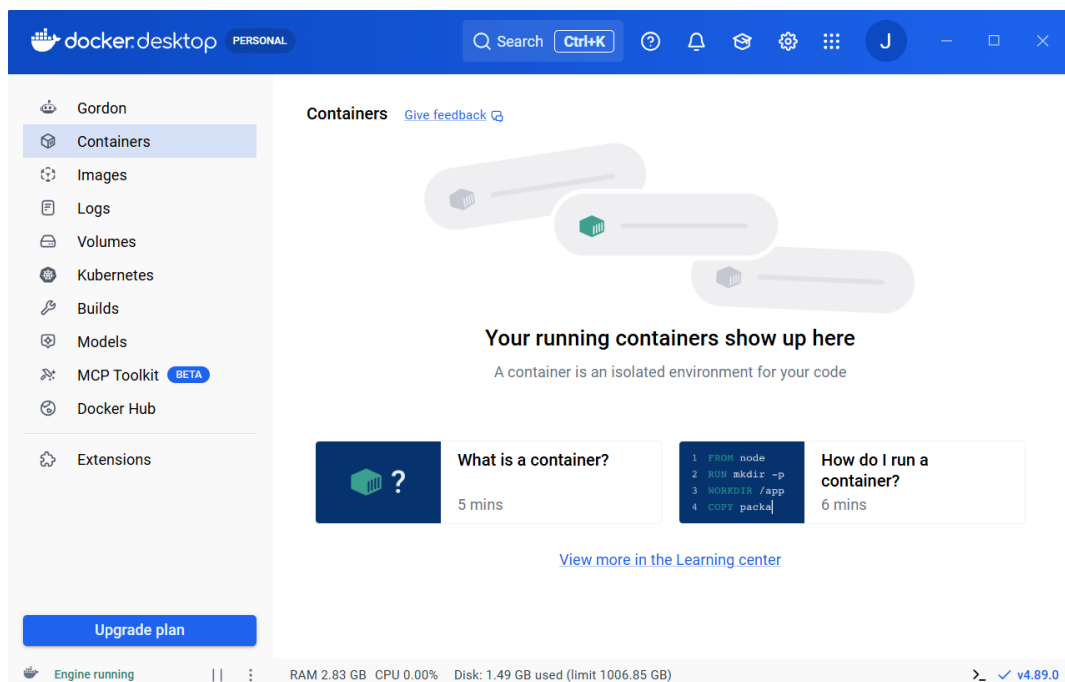
Oharra: Docker Desktop-ek **Docker Engine** eta **Docker Compose** barne ditu. Ez da Docker Compose aparte instalatu behar.

1.5. Instalazioa

1. Exekutatu deskargatutako instalatzailea.
2. Onartu lizentzia.
3. Mantendu lehenetsitako aukerak:
 - Use WSL2 instead of Hyper-V (gomendatua).
 - Add shortcut to desktop (aukerakoa).
4. Sakatu **Install**.
5. Instalazioa amaitutakoan, sakatu **Finish** eta berrabiarazi ordenagailua eskatzen badu.

1.6. Docker lehen aldiz abiaraztea

1. Ireki **Docker Desktop**.
2. Baldintzak onartu
3. Itxaron motorra abiarazi arte, hau da, beheko aldean **Engine running** mezua agertu arte.



1.7. Docker instalazioa egiaztatzea

Ireki **PowerShell** edo **Komando Leihoa (CMD)** eta exekutatu:

```
docker --version
```

Espero den emaitza: *Docker version 29.x.x*

Ondoren exekutatu:

```
docker compose version
```

Espero den emaitza: *Docker Compose version v5.x.x*

Eta azkenik:

```
docker run hello-world
```

Dena ondo badago, Docker-ek irudi ofiziala deskargatuko du eta mezu hau agertuko da: *Hello from Docker!*

Horrek adierazten du instalazioa behar bezala egin dela.

1.8. Arazo ohikoenak

- **“WSL 2 installation is incomplete”**

Konponbidea:

```
docker --version
```

Ondoren berrabiarazi Docker Desktop.

- **“Virtualization is disabled”**

Konponbidea: BIOS/UEFIra sartu eta birtualizazioa gaitu.

- **Docker ez da abiarazten**

Exekutatu:

```
wsl --shutdown
```

Ondoren ireki berriro Docker Desktop.

- **Linux banaketa ez da agertzen**

Exekutatu:

```
wsl --install edo wsl --list --online
```

Eta instalatu banaketa bat.

2. Python 3 instalatzea

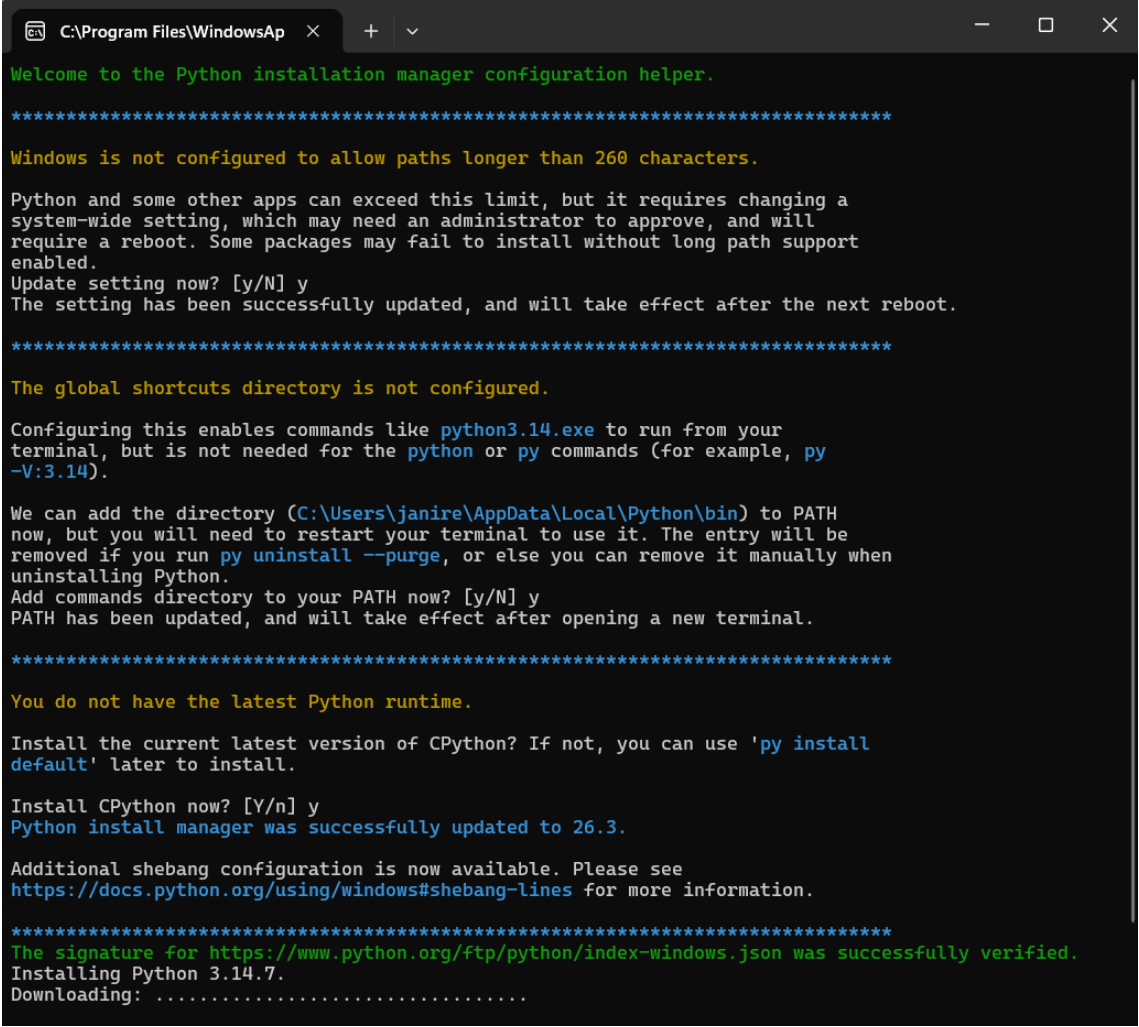
2.1. Deskargatu

Joan webgune ofizialera: <https://www.python.org/downloads/>

Deskargatu azken bertsio egonkorra.

2.2. Instalazioa

Exekutatu instalatzailea. **Install Now**

A screenshot of a Windows command prompt window titled 'C:\Program Files\WindowsAp'. The window displays the 'Python installation manager configuration helper' interface. It shows several configuration steps: 1. A warning about Windows not being configured for paths longer than 260 characters, with a confirmation to update the setting. 2. A warning about the global shortcuts directory not being configured, with a confirmation to add the directory to the PATH. 3. A confirmation to install the current latest version of CPython (26.3). 4. A confirmation to install CPython now. 5. A message that the signature for the Python installation was successfully verified. The window is dark-themed with green and yellow text for warnings and green text for success messages.

```
C:\Program Files\WindowsAp Welcome to the Python installation manager configuration helper.

*****
Windows is not configured to allow paths longer than 260 characters.

Python and some other apps can exceed this limit, but it requires changing a
system-wide setting, which may need an administrator to approve, and will
require a reboot. Some packages may fail to install without long path support
enabled.
Update setting now? [y/N] y
The setting has been successfully updated, and will take effect after the next reboot.

*****
The global shortcuts directory is not configured.

Configuring this enables commands like python3.14.exe to run from your
terminal, but is not needed for the python or py commands (for example, py
-V:3.14).

We can add the directory (C:\Users\janire\AppData\Local\Python\bin) to PATH
now, but you will need to restart your terminal to use it. The entry will be
removed if you run py uninstall --purge, or else you can remove it manually when
uninstalling Python.
Add commands directory to your PATH now? [y/N] y
PATH has been updated, and will take effect after opening a new terminal.

*****
You do not have the latest Python runtime.

Install the current latest version of CPython? If not, you can use 'py install
default' later to install.

Install CPython now? [Y/n] y
Python install manager was successfully updated to 26.3.

Additional shebang configuration is now available. Please see
https://docs.python.org/using/windows#shebang-lines for more information.

*****
The signature for https://www.python.org/ftp/python/index-windows.json was successfully verified.
Installing Python 3.14.7.
Downloading: .....
```

```

*****
The signature for https://www.python.org/ftp/python/index-windows.json was successfully
Installing Python 3.14.7.
Downloading: ..... ✓
Extracting: ..... ✓
To see all available commands, run 'py help'
*****

*****

Configuration checks completed.

To run these checks again, launch Python install manager from your Start menu,
or py install --configure from the terminal.

*****

Usage:
py <regular Python options>
    Launch the default runtime with specified options. This
    is the equivalent of the python command.
py -V:<TAG>
    Launch runtime identified by <TAG>, which should
    include the company name if not PythonCore. Regular
    Python options may follow this option.
py -3<VERSION>
    Equivalent to -V:PythonCore\3<VERSION>. The version
    must begin with the digit 3, platform overrides are
    permitted, and regular Python options may follow. py -3
    is the equivalent of the python3 command.
py exec <any of the above>
    Equivalent to any of the above launch options, and the
    requested runtime will be installed if needed.
py help [<CMD>]
    Show help for Python installation manager commands
py install <TAG>
    Download new Python runtimes, or pass --update to
    update existing installs.
py list [<FILTER>]
    Show installed Python runtimes, optionally filtering by
    <FILTER>.
py uninstall <TAG>
    Remove one or more runtimes from your machine. Pass
    --purge to clean up all runtimes and cached files.

Find additional information at https://docs.python.org/using/windows.
View online help? [y/N] |

```

2.3. Python instalazioa egiaztatzea

PowerShell-en idatzi:

```
python -version
```

Espero den emaitza:

Python 3.14.x

2.4. Pip egiaztatzea

PowerShell-en exekutatu:

```
pip -version
```

Espero den emaitza:

pip 26.x.x...

3. Docker Compose probatzea

3.1. Sortu karpeta bat

```
mkdir docker-proba  
cd docker-proba
```

3.2. Sortu compose.yaml fitxategia

```
notepad compose.yaml
```

Edukia:

```
services:  
  web:  
    image: nginx  
    ports:  
      - "8080:80"
```

3.3. Abiarazi zerbitzua

```
docker compose up
```

3.4. Egiaztatu

Ireki nabigatzailea eta joan helbide honetara:

```
http://localhost:8080
```

Nginx-en hasierako orria agertu behar da.

3.5. Amaitu

Terminalean:

```
Ctrl + C
```

Ondoren:

```
docker compose down
```